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Examining the effect of task constraints on the emergence of creative action in young elite football players by using a method combining expert judgement and frequency count

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ABSTRACT

In adult football, small-sided games are associated with increased action variability and suggested to promote more creative actions compared to regular 11v11 formats. This aligns with predictions from an ecological approach to perception and action that creative actions emerge in environments that grant variability in action, instead of being an expression of the individual player's ability to generate ideas. To further evidence for this prediction, the current study aimed to expand this observation to elite youth football players. To this end, the number of different and creative actions in 4v4 small-sided game and a 11v11 regular-sided game among 10- to 12-year-old elite football players were examined. We analyzed a total of 7922 actions, which were categorized for type and creativity. Based on a subset of these actions, a panel of elite football coaches judged action types occurring below 0.5% as significantly more creative than more frequent action types. Hence, we used an occurrence of 0.5% as threshold to distinguish creative actions from non-creative actions. The results showed that the total number of actions, the number of different action types, the number creative actions and the number of different creative action types was significantly higher for the small-sided game format than the regular-sided game. In conclusion, this study confirms that in elite youth football, small-sided games induce a more variable and creative action repertoire. This shows that practitioners can design learning environments that promote the emergence of creative actions.

1. Introduction

Although sometimes elusive, creativity has long been a popular topic of study in general psychology (Sternberg & Lubart, 1999). Over the past two decades, the importance of creativity has been increasingly recognized in sport science as well (Fardilha & Allen, 2019). For instance, in international football tournaments, more successful teams show higher creativity in their actions, and importantly, the degree of creativity in the final actions leading into a goal is associated with match success (Kempe & Memmert, 2018). In the *Handbook of Creativity*, Sternberg and Lubart (1999), captured creativity as an individual ability "to produce work that is both novel (i.e., original, unexpected) and appropriate (i.e., useful)" (Sternberg & Lubart, 1999, p. 3). Following this, researchers in sports, and increasingly practitioners as well, have considered creativity as a process of divergent thinking, where creative actions are underpinned by individual athletes' ability to generate as many different actions or solutions as possible (Memmert, 2015). Accordingly, creativity can be enhanced by stimulating divergent thinking. Contrary to this cognitively-oriented top-down approach, in which creative actions are merely an expression of an athlete's thinking, others have argued that creative actions foremostly emerge in the (inter)actions of the athlete in the sporting environment (Orth et al., 2017; Zahno & van der Kamp, 2022). Indeed, according to proponents of the ecological approach, new, original actions arise in an athlete's attempt to adapt to the dynamic environment rather than from an isolated process of idea generation (Glăveanu, 2013; Orth et al., 2017; Westmeyer, 1998; Withagen & van der Kamp, 2018). Skilled athletes have a larger, more variable, degenerate motor repertoire, which allows them to adapt more flexibly in a situation, that is, they can achieve the same solution with different actions. It stands to reason that when capable of producing

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more variable, degenerate actions, there is an increased likelihood that some of those actions are comparatively unique or rare, and thus recognized as creative (Orth et al., 2017; Simonton, 2003; Zahno & van der Kamp, 2022). This conceptualization holds that creative actions are not caused by individual abilities, but emerge in the athlete's to search to adapt to task and environmental constraints. For instance, Dick Fosbury famously won the 1968 Olympic gold medal in high jumping by demonstrating a new technique, known as the Fosbury flop technique. This new, original and functional technique, in which he landed on his shoulders, was only possible after the introduction of the thick and soft crash mat. Following this, practitioners could design learning environments with task constraints that invite variable and degenerate behavior and, therefore, increase the chance for creative actions to emerge (Hristovski et al, 2011; Orth et al., 2017). Although this might be essential for talent development, evidence in this framework is scarce and limited to adults.

Indeed, research has started to show that manipulating tasks and environmental constraints evokes creative actions. In football, an example is the use of small-sided games (SSGs). In small-sided game formats the number of players is typically decreased, the size of the playing field reduced and/or the rules of the games adjusted to create more dynamic, variable practice environments for high intensity training. SSGs are often used in training to improve players' combined technical, tactical and physiological skills and capacities (Drust et al., 2000; Owen et al., 2004). Compared to regular-sided 11v11 matches (RSGs), SSGs show a more dynamic and less structured environment, which is suggested to enhance creativity (Bowers et al., 2014; Greco et al., 2010; Memmert et al., 2010). For instance, reducing the number of players on a smaller playing field resulted in a significant increase in the number of ball contacts and actions (e.g., passing, shooting, and dribbling) performed by individual players (Owen et al., 2004). In other words, SSGs provide players more opportunities to perform actions with the ball, resulting in players demonstrating a larger and more variable action repertoire. It is therefore expected that SSGs promote the emergence of creative actions (Hristovski et al., 2011; Orth et al., 2017).

Initial evidence for this proposal is reported in a study that examined the emergence of variable and creative actions in different SSGs and RSGs (Caso & van der Kamp, 2020). Elite adult players were found to demonstrate a richer (i.e., more different actions) and more creative action repertoire in small SSG-formats (i.e., 5v5, 6v6 and 7v7) than in regular formats (i.e., 11v11). Hence, SSGs may provide promising practice conditions for increasing young players' action repertoires, and accordingly, their adaptive flexibility and creativity, enabling them to surprise opponents with new and unexpected actions (Durand-Bush & Salmela, 2002). As such, stimulating the acquisition of sport specific skills and degenerate behavior can be essential for talent development. However, there is currently no empirical evidence confirming that the dynamics of SSGs also increase the -creative- action repertoire of young football players.

Before assessing the effects of SSGs on the emergence of creative actions in youth players, it is important to critically reflect on the methods used to identify creative actions in the study by Caso and van der Kamp (2020, see also Orangi et al., 2021). They performed a notational video analysis and categorized all actions performed by each player during the SSGs and football matches. A frequency count was used to decide if an action was creative, that is, if it was both original and functional. An action was considered original if no more than 5% of the players performed that action, and functional if it was performed adequately, that is, achieved the intended goal. This purely quantitative approach solely defines originality on frequency count (see also Simonton, 2003). This notably diverges from other approaches in sport science, where creative actions are identified based on intersubjective ratings of experts (e.g., Hendry et al., 2018; Kempe & Memmert, 2018; Memmert, 2010; Zahno & Hossner, 2022). A quantitative approach diminishes biases from the individual experiences or opinions of the expert judge, or a fixed idea about what is and is not supposed to be creative (Bergkamp et al., 2019; Roberts et al., 2019). Yet, there are three issues with the quantitative approach as used by Caso and van der Kamp (2020). First, by taking the frequency of players performing a particular action to define originality, it implicitly presumes that creativity is an attribute of the individual player. Instead, it seems theoretically more appropriate to use the frequency of the action type itself for setting the criterion for originality, irrespective of the number of players producing the action. Second, the chosen threshold of 5% was arbitrary, as would have been any other chosen numerical threshold. That is, a quantitative approach may return actions that nobody would consider creative, because it is insensitive to the broader conventions and habits of a behavior setting (Barker, 1968). Lastly, because of the small number of creative actions that they identified, Caso and van der Kamp (2020) decided not to statistically verify that SSGs indeed had increased the number of creative actions. In the current study, therefore, we combined both methods. We used the experts' judgements to define the frequency threshold for deciding which action is creative.

The purpose of the current study is to examine if a richer and more creative action repertoire can be promoted among elite youth football players between 10 and 12 years of age by implementing SSGs that are presumed to invite a larger variability of actions. To this end, we compare the number of different and creative actions for 4v4-SSGs with regular 11v11 matches. To distinguish creative actions, we use a numerical frequency threshold that is derived from judgments of creativity by expert coaches. Based on the observations in elite adult players (Caso & van der Kamp, 2020), we hypothesize that 4v4-SSG format results in more actions, more different actions types, more creative actions and more different creative actions types compared to regular 11v11 matches.

2. Method

2.1. Participants

Participants included 240 youth football players between 10 and 12 years of age ($M = 11.1$, $SD = .12$) who were affiliated with an elite football academy of a Dutch premier league club. Video recordings from 49 SSGs and 4 regular 11v11 matches were included. The unit of analysis was the game; therefore the 4 regular matches were divided in 43 equal parts of 6 min, which was approximately the average duration of the SSGs. Caso and van der Kamp (2020) reported an effect size $d = .68$ for the number of different actions types (i.e., they did not report statistics for the number of creative actions). Based on this an a priori power analysis ($\alpha = 0.05$, $1\text{-}\beta = 0.80$, $d = .68$) indicated a minimum sample size of 37. 4v4-SSG amounted to 4846 actions from 293 min actions and 3076 actions from 260 min 11v11 regular matches. Team composition differed within and across SSGs and matches but consisted of players of the same age and skill. The local ethics committee had approved the study procedures, and parents or guardians of the participants provided written informed consent. Finally, to define the frequency threshold for creativity four expert coaches with on average 11.4 years of experience ($SD = 3.5$) were recruited as judges. At the time of study, the coaches were each employed as a professional coach at the same Dutch premier league club as from the youth players were recruited. Two were head coach of a youth team (i.e., U12 & U13), the third was also coach of a youth team (i.e., <U14) and the coordinator of all youth teams, and the fourth was the head coach of the first women team. They also provided written informed consent.

2.2. Design

The 4v4-SSG consisted of training games among members of the same team. A total of 7 teams were included for SSGs. All 49 SSGs were set to 40m length and 20m width and lasted between 6 and 9 min. Two goals were placed on the short side of the field, no goalkeepers were allowed at the goal. During the SSGs no throw-ins were taken; in case the ball left the field, the players were supposed to continue playing by passing the ball back into the area of play. The 11v11-RSG consisted of friendly games against teams of different clubs playing at the same competitive level. Two friendly games were played within the club between two teams of a different age. In total 7 teams were included for RSGs. The 11v11 match environment was according to the official Royal Dutch Football Association (KNVB) game regulations. The field consisted of 105m length by 68m width, and the matches took between 40 and 60 min. The matches were played on an artificial grass pitch. Both the 4v4 and 11v11 games were recorded by a video analyst employed at the football club.

2.3. Data and statistical analysis

2.3.1. Coding of actions

The video footages were analyzed by two experienced video analysts with the program BORIS (Behavioral Observation Research Interactive Software, v. 7.9.8), which allowed tracking of the actions throughout the games. The game analysis included all ball actions of each player, including goalkeepers' actions without their hands during 11v11 matches. The unit of analysis was the game, this allowed to consider the actions to emerge from the environment, rather than being produced by an individual player. The categorization of type of actions in Caso and van der Kamp (2020) served as the starting point for the coding instrument. However, it was modified to incorporate all the actions performed in the recorded video-footage. In addition, we restricted analysis to actions of players that were in possession of the ball, meaning that actions, runs, and other behaviors without the ball were not coded. Furthermore, the first touch and dribbling actions passing opponents simply by increasing pace were also excluded for analysis. This resulted in list of 32 action types (Appendix 1). Interobserver reliability was determined for categorization of actions during 30 min of SSG and 25 min of one 11v11 game. A substantial interobserver agreement was found for type of action between the two observers (Cohen's $\kappa = .707$, $p < .001$) (Landis & Koch, 1977). Each action was also categorized as either functional or non-functional. An action was functional if it fulfilled the purpose for that type of action and resulted in ball possession being maintained or a goal scored (e.g., successfully performing a scissor action without any opponent near or reacting to it was non-functional). A substantial interobserver reliability was also found for functionality (Cohen's $\kappa = .638$, $p < .001$).

2.4. Determining the creativity frequency threshold

The subsequent data processing consisted of four steps. First, the relative frequency for each action type for both the 4v4-SSG and 11v11-RSG was calculated. Second, the four members of the expert panel judged a subset of 65 actions presented in separate videoclips with respect to creativity, using a procedure based on the Consensual Assessment Technique (CAT, (Amabile et al., 1996)). The CAT assumes that creative actions can be identified if experts in the field independently agree that a product or response is creative (Amabile et al., 1996). The CAT is suggested to be the best judgement tool and the golden standard to examine creativity (Baer, 2015; Kaufman et al., 2008; Zahno & Hossner, 2022). Following this method, three procedural steps were pursued: (1) expert coaches were independently asked to judge the creativity of each of the presented actions, (2) the expert coaches made their judgments within situations that are related to their domain of expertise, (3) the actions were judged relative to one another. This procedure was identical for the SSGs and RSGs. Amabile et al. (1996) added that the experts should not be trained or instructed before the assessment. In the current study, however, the experts were briefly informed that creativity should be understood as an action that is both functional and original, but without further defining these terms. They were asked to give a holistic judgement with respect to the creativity of the presented action and rated this on a 5-point Likert scale for creativity, 1 'the action is not at all creative' and 5 'the action is very creative'. To obtain the experts' judgements, video-footage of 65 actions were prepared in separate videoclips that included events between 3 and 5 s before and after the target action. This allowed the judges to take into account the game situation and the outcome of the action. Each clip started as a frozen video image with a red circle around the player of interest that would make the target action. Before the actual assessment, five clips were shown for familiarization. The remaining 60 clips included 10 functional actions with a relative frequency of 0.5% (i.e., =< 0.5%), 10 functional actions with a relative frequency of 1% (i.e., >0.5% and =< 1.0%), 10 functional actions with a relative frequency of 3% (i.e., >1.0% and =< 3.0%), and 10 functional actions with a relative frequency of 10% (i.e., >3.0% and =< 10.0%). Additionally, 10 non-original functional actions with a relative frequency of more than 10% and 10 non-functional actions were included. To establish the frequency threshold for an action to be creative, two requirements had to be met: the average expert creativity rating for actions within a relative frequency category (i.e., 0.5%, 1%, 3% and 10%) should be significantly higher than the ratings for both the non-original and the non-functional action category. This was tested with a Kruskall-Wallis H test, with a one-tailed Bonferroni-corrected post-hoc comparisons. All functional actions with lower relative frequency than this frequency threshold were defined as creative.

2.5. Comparing environments: 4v4-SSG and 11v11-RSG

The 49 intervals of 6 min for SSG and 43 intervals for RSG were analyzed for the total number of actions, the number of different action types, the number of creative actions and the number of different creative action types. A Shapiro-Wilk test was carried out to confirm normal distribution and Levene's test was used to assess equal variance, which was violated. Accordingly, Mann-Whitney tests were conducted to compare the dependent variables between the 4v4-SSGs and the 11v11-RSGs.

3. Results

3.1. Total number of actions

A total of 7922 were identified, of which 5750 were functional (73%) and 2172 were non-functional (27%). There were significantly more actions for the 4v4-SSG compared to the 11v11-RSG, $U = 234.5$, $p < .001$, $r = .70$ (Table 1). Indeed, this involved both significantly more functional, $U = 194.5$, $p < .001$, $r = .37$, and non-functional actions, $U = 784.5$, $p = .035$, $r = .22$, in the SSG compared to the RSG.

Table 1
The median number of actions (IQR) in the 4v4-SSG and the 11v11-RSG in intervals of 6 min.

Actions	4v4	11v11
Total	102.0 (24.0)	69.0 (17.0)
Functional	75.0 (19.5)	49.0 (14.0)
Non-functional	25.0 (9.0)	21.0 (10.0)

Table 2
The median number of action types (IQR) in the 4v4-SSG and the 11v11-RSG in intervals of 6 min.

Action types	4v4	11v11
Total	14.0 (3.0)	13.0 (3.0)
Functional	12.0 (3.0)	11.0 (2.0)
Non-functional	8.0 (3.5)	8.0 (3.0)

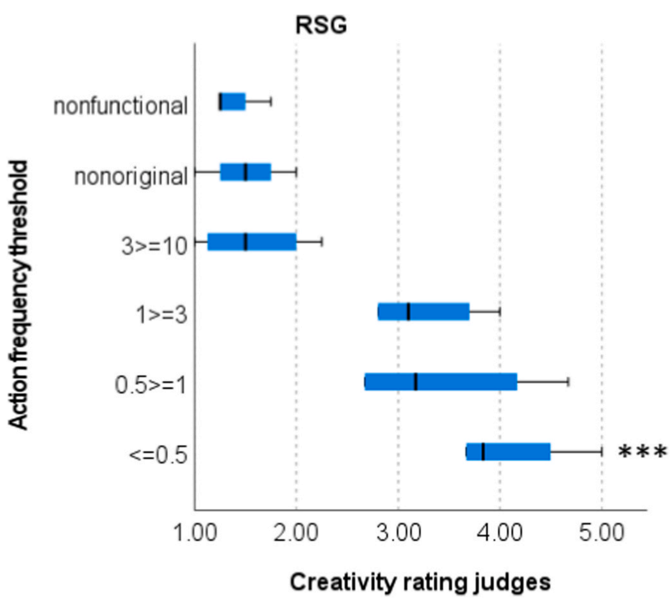

Fig. 1. Boxplot for creativity rating of judges for 11v11 regular-sided game as a function of action frequency threshold. ***Significant effect with non-functional category, non-original and 10% threshold (p < .05).

3.2. Number of different action types

Table 2 shows a significant effect for environment with respect to the total number of different action types, U = 744.0, $p = .014$, $r = .26$. The variation of functional action types was significantly higher for SSG compared to RSG, U = 609.5, $p < .001$, $r = .37$. No difference in variation of non-functional action types was found between SSG and RSG, U = 1040.5, $p = .92$ (Table 2).

3.3. Creativity threshold

A multiple rater, absolute agreement, two-way mixed-effects model was conducted to assess interobserver reliability for the expert judges, a good and significant agreement was found for the creativity rating (ICC = .84, p < .001) (Koo & Li, 2016). For the RSG, a significant effect for frequency threshold was found for the creativity rating, $H(5) = 18.2$, $p = .001$, $\eta_p^2 = .73$. One-tailed post hoc comparisons with Bonferroni correction indicated that the judges rated the actions with relative frequency of 0.5% significantly more creative than the non-functional actions ($p = .018$, $\eta_p^2 = .64$), more creative than actions in the non-original category ($p = .037$, $\eta_p^2 = .59$) and actions within the relative frequency of 10% threshold ($p = .037$, $\eta_p^2 = .59$). No other differences were revealed (Figure 1). This points to the 0.5% frequency being the threshold for action types to be identified as creative within RSG.

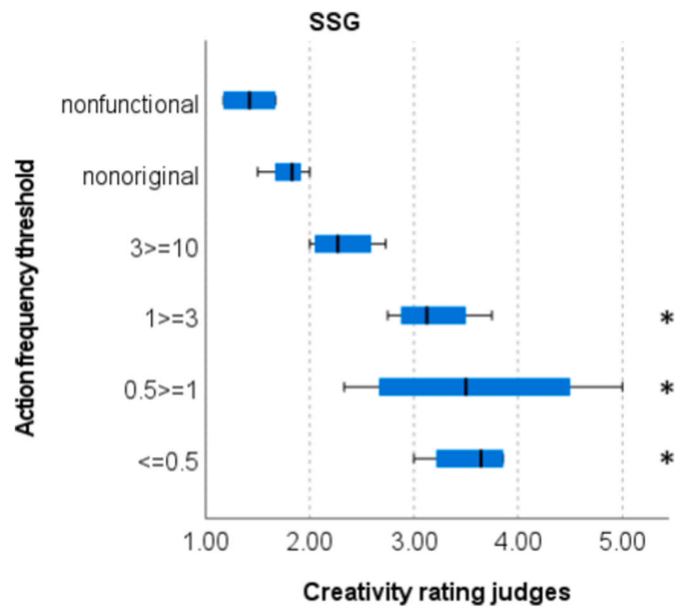

Fig. 2. Boxplot creativity rating indicated of judges for 4v4 small-sided game as a function of action frequency threshold. *Significant effect with non-functional category (p < .05).

For the SSG, a significant effect for frequency threshold was found for the creativity rating, $H(5) = 19.2$, $p = .002$, $\eta_p^2 = .79$. One-tailed post hoc comparisons with Bonferroni correction indicated that the judges rated actions with a relative frequency of 0.5% significantly more creative than actions within the non-functional category ($p = .007$, $\eta_p^2 = .70$). Also, a non-significant difference was found compared to non-original actions ($p = .055$). Finally, actions within a relative frequency of 1% and 3% were also rated significantly more creative than the non-functional actions (respectively, $p = .014$, $\eta_p^2 = .66$ and $p = .038$, $\eta_p^2 = .60$), but differences with the non-original actions were not significant (respectively, $p = .098$ and $p = .220$). No other differences were present (Figure 2). The requirements were not completely met, but consistent with the RSG, this suggests the 0.5% criterion to be the most appropriate for defining the frequency threshold for creativity.

3.4. Creative actions

Table 3 shows that the 4v4-SSG induces more creative actions than the 11v11-RSG, with the 0.5% frequency threshold. A Mann-Whitney test confirmed this, U = 704.0, $p = .004$, $r = .30$. In addition, the SSG showed more creative action types than the RSG, U = 733.5, $p = .008$, $r = .28$.

Table 3
The median number of creative actions and creative action types (IQR) for 4v4-SSG and 11v11-RSG in intervals of 6 min.

	4v4	11v11
Creative actions	2.0 (2.0)	1.0 (1.0)
Creative action type	1.0 (1.0)	1.0 (1.0)

4. Discussion

Players in team sports perform in very dynamic environments that ask a manifold of actions to satisfy the task constraints. A large action repertoire is not only a necessity to adapt to such dynamic performance environments, but is also thought to increase the likelihood that rare, creative actions occur, which can surprise opponents (Glăveanu, 2013; Hristovski et al., 2011; Orth et al., 2017; Rasmussen et al., 2019; Zahno & van der Kamp, 2022). Recently, it was reported that manipulation of task constraints related to the number of players and field size (i.e., small-sided games, SSG), increase the variability and creativity of actions in elite adult players (Caso & van der Kamp, 2020). Consequently, an environment that encourage to a more extensive exploration of the task might also induce more variable and creative actions in elite youth football players. This might be essential for talent development, therefore, the current study examined whether manipulating task constraints for small-sided games (4v4), also induce more variable and creative actions in elite youth football players.

The results showed that 4v4-SSGs invited more functional actions compared with the 11v11-RSG matches. This is in agreement with previous theoretical and empirical studies that small-sided football games with less players and smaller space induce more actions (Davids et al., 2013; Owen et al., 2004). We anticipated that SSGs also induce a larger variety of actions. Indeed, the current observations indicated more variability in actions or a broader action repertoire for the 4v4-SSGs compared with the regular 11v11 matches. This corroborates recent observations that elite adult players show a more variable action repertoire in SSGs (Caso & van der Kamp, 2020). Among the elite adult players, this was also associated with the emergence of more creative actions. Again, this was confirmed for the talented youth football players. Using a 0.5% frequency threshold for creativity, we found more creative actions and more creative action types for SSGs compared to regular 11v11 matches. Specifically, SSGs (4v4) increase the likelihood for an individual player to perform a creative action with factor 4.2 compared to a RSG (11v11). In sum, also with talented youth players the constraints emerging within SSGs lead to higher action variability and more creative actions, presumably because the SSGs invite more exploratory behavior. The current study confirms earlier observation in adults (Caso & van der Kamp, 2020), but also extends these to youth football. This indicates the potential for SSGs in the acquisition of a broad and creative action repertoire in talent development, which should be examined in longitudinal studies.

The identification and analysis of creative actions in current study deviated from previous purely quantitative assessments (e.g., Caso & van der Kamp, 2020; Orangi et al., 2021) and with the regular intersubjective assessments from expert judges (Hendry et al., 2018; Kempe & Memmert, 2018; Memmert, 2010; Zahno & Hossner, 2022). The quantitative approach defined an action as creative based only on a numerical frequency count. If 5% or fewer players performed an action and it was also functional, then it was defined as creative (Caso & van der Kamp, 2020; Orangi et al., 2021). This approach not only (tacitly) assumed that creativity is an attribute of the individual player, but also used a largely arbitrary frequency threshold. In addition, within a purely quantitative approach expert knowledge about the game is not considered. By contrast, even though an expert judgement is holistic and in all likelihood biased by idiosyncrasies (Bergkamp et al., 2019; Roberts et al., 2019), it is sensitive to situation-specific habits and conventions of behavior setting or social practices (Barker, 1968) that are neglected in a purely quantitative approach. Hence, in the current study, we combined the two methods by asking expert judges a holistic rating of creativity of an action, which will be grounded in what is implicitly sensed as creative in their domain of expertise, and used these ratings to define the frequency threshold for identifying an action type as creative. The inter-observer agreement between the experts' rating was good, which exemplifies a communis opinion that presumably reflects the shared football habits, conventions and opinions among the coaches. The experts' judgments defined the frequency threshold for the subsequent quantitative identification of creative actions. Consequently, the inter-subjective expert judgments determined the maximum frequency of an action to be defined creative, instead of defining what actions should be considered creative.

We used this method to define separate frequency thresholds for the two environments. The resultant thresholds were the same, and based on the 0.5% threshold we confirmed that SSGs invite more creative actions than regular 11v11 games in talented young football players. Nevertheless, it raises the issue on the specificity of the frequency thresholds. It stands to reason, especially within the theoretical framework of ecological psychology, that the frequency threshold for creativity depends on the interacting individual, task and enviroment constraints, that is, type of game, players' age and skill level, performance context and so on. The generality vs specificty of the thresholds is an important avenue for further research, also from the perspective of designing practice environments for talent development. Practice within SSGs may not necessarily lead to increased emergence of creative actions in RSGs, especially if task constraints are critically different between the game formats. For example, tactical constraints in a SSG and RSG with respect to player positions and field roles are non-identical, which may indeed invite different actions. Hence, each environment, even within the same sport, may induce actions that are considered creative in one but not in another context. Consequently, it is perhaps more fruitful for talent development to design environments that challenge adaptivity and increase variability in the action repertoire, rather than creative actions per se.

In this respect, it is important to notice a few other limitations of the current study. First, two 11v11 games were played with teams of different age groups (i.e., U11 vs U13 an U12 vs U13). Older players are more technical developed (Huijgen et al., 2013) and tend to have more diverse tactical decisions compared with younger players (Barnabe et al., 2016; Clemente et al., 2020; Olthof et al., 2015). This likely presents different task constraints. For example, older players are likely capable of a broader action repertoire and explore more task-related space, possibly leading to more creative actions compared to younger players (Orth et al., 2017). Additionally, Santos et al. (2023) have recently suggested that a larger number of creative opponent players constrains team-tactical creative behaviors of the opponent, albeit that they did not actually observe that it affected the emergence of individual creative actions. Nevertheless, too demanding or too easy scenarios might not lead to technical and tactical explorative behavior (Torrents et al., 2016, 2021). It, therefore, seems appropriate for talent development to seek for balanced constraints for age and skill level between contending teams too increase creative behavior. Second, the current study only examined creative with-ball action categories, but obviously, football and thus creative actions in football consists of more than technical ball skills, it also includes actions without ball and team-tactical actions (Caso & van der Kamp, 2020; Memmert, 2015; Memmert & Roth, 2007; Santos et al., 2023). We do not think that including them would lead to fundamentally different conclusions, but it is recommended for future studies to at least try to describe the strategical and tactical context in which the teams and players operate. This may also be fruitful in further aligning the purely quantitative method and the more idiosyncratic expert judgements.

We consider two practical implications of the current study as important: (1) for talent development, football coaches should not primarily seek to nurture creative players, but rather should promote to explore and broaden the action repertoire, which as a corollary will increase the likelihood to emerge creative behavior. Designing training environments in which pertinent task constraints are manipulated, such as the smaller field size and/or reduced number of player in SSG is recommended; (2) for talent identification and development, video analysts can monitor the frequency and variability of the action repertoire, and in consultation with coaches and/or scouts determine benchmarks for creative actions. Although current findings indicated the same frequency threshold for SSG and RSG, this benchmark may be re-assessed for different age, game-formats and so forth, depending on the imposed constraints.

In conclusion, we confirm previous observations in adult football that small-sided games compared to regular sides games invite a richer action repertoire and more creative actions, and extend these findings to 10- to 12-year-old elite football players. This shows that creative actions emerge in environments that grant variability in action, instead of being an expression of the individual player's ability to generate ideas. Our results are based on the development of a new method for identifying creative actions that combines a quantitative frequency count with expert judgement. These findings have implications both for talent identification and development, although it is important to address the generality versus specificity of the used method before a full-scale implementation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

Appendix 1

Action type	Definition
Pass/shot with inside foot	Player in possession sends the ball to a teammate with inside of the foot.
Pass/kick outside foot	Player in possession sends the ball to teammate with outside of the foot/kicks the ball to score a goal.
Pass/kick volley	Player in possession sends the ball to teammate with volley/volleys the ball to score a goal.
Chip pass/kick	Player in possession sends the ball to teammate by chipping the ball/chips the ball to score a goal.
Pass with chest	Player in possession sends the ball to teammate by using the chest.
Pass with foot heel	Player in possession sends the ball to teammate with the heel of the foot.
Pass with knee	Player in possession sends the ball to teammate with the knee.
Rabona	Player strikes the ball when it is on the outside of your standing leg.
Header passing/shooting	Player heads the ball to teammate or scores a goal.
Scissor (step over)	Player steps with one foot wide over the ball, pushes ball away with other foot towards movement direction.
Feint	Player make a body movement towards one direction and moves towards other direction
Drag Back	Players drags the ball backwards.
Stop, spin, go	Players stops the ball with foot, spins towards another direction and moves on.
Double Scissor (double step over)	Player steps with two footsteps wide over the ball, thereafter, pushes ball away with first foot.
Ronaldo chop	Players strikes ball with the inside of the foot behind the standing leg.
Zinedine Zidane (Roulette)	Player puts foot on top of the ball then rolls it back. At the same time the body spins and the ball rolls with the other foot towards movement direction.
Turning Cruyff	Player drags the ball back through the legs.
Flip Flap (Ronaldinho)	Players drives the ball with the outside of the foot thereafter changes direction immediately by bringing the ball with the inside of the foot towards other side of the body.
Fake shot	Player moves like a shot and slides the ball to the side at the moment of ball contact.
Split	Player passes opponent by playing the ball beside the opponent and runs around the opponent the other side.
Drag	Player makes a long drag movement with the inside of the foot which causes threat to one side and passes the opponent on the other side.
Dummy	Players steps over the ball while it was passed from a teammate.
Forward step over (Rivelino)	Player steps over the ball thereafter takes the ball in the opposite direction with the same foot.
Chip	Player slightly touches the ball which causes the ball to move up outside the range of the opponent.
Chop	Player strikes the ball with the side of the foot and changes the direction of the ball with an angle of 45–90°
La Croqueta (Iniesta move)	Player moves the ball with the inside of the foot with an angle of 45–60° to the other foot and takes it forward.
Toe-poke pass/shot	Player in possession passes the ball or shoots the ball with the front of the foot.
Backheel	Player in possession drags the ball to the back with one foot and plays it behind the back with the heel of this foot.
Lob	Player plays the ball over the head of the opponent to pass by or to score a goal
Flap flip	Players drives the ball with the inside of the foot thereafter changes direction immediately by bringing the ball with the outside of the foot towards other side of the body.
Outside Turn	Player lets the ball slide over the outside of his foot to turn around the opponent.
Nutmeg	Player intentionally passes the ball through opponent's legs to run past him or pass the ball to their teammate.

References

Amabile, T. M., Amabile, T. M., Collins, M. A., Conti, R., Phillips, E., Picariello, M., Ruscio, J., & Whitney, D. (1996). *Creativity in context.* Routledge. https://doi.org/10.4324/9780429501234

Baer, J. (2015). The importance of domain-specific expertise in creativity. *Roeper Review, 37*, 165–178. https://doi.org/10.1080/02783193.2015.1047480

Barker, R. G. (1968). *Ecological psychology: Concepts and methods for studying the environment of human behavior.* Stanford University Press.

Barnabé, L., Volossovitch, A., Duarte, R., Ferreira, A. P., & Davids, K. (2016). Age-related effects of practice experience on collective behaviours of football players in small-sided games. *Human Movement Science, 48*, 74–81, 10.1016/j.humov.2016.04.007.

Bergkamp, T. L. G., Niessen, A. S. M., den Hartigh Ruud, J. R., Frencken, W. G. P., & Meijer, R. R. (2019). Methodological issues in soccer talent identification research. *Sports Medicine, 49*(9), 1317–1335. https://doi.org/10.1007/s40279-019-01113-w

Bowers, M. T., Green, B. C., Hemme, F., & Chalip, L. (2014). Assessing the relationship between youth sport participation settings and creativity in adulthood. *Creativity Research Journal, 26*, 314–327. https://doi.org/10.1080/10400419.2014.929420

Caso, S., & van der Kamp, J. (2020). Variability and creativity in small-sided conditioned games among elite soccer players. *Psychology of Sport and Exercise, 48*, Article 101645. https://doi.org/10.1016/j.psychsport.2019.101645

Clemente, F. M., Castillo, D., & Los Arcos, A. (2020). Tactical analysis according to age-level groups during a 4 vs. 4 plus goalkeepers small-sided game. *International Journal of Environmental Research and Public Health, 17*(5). https://doi.org/10.3390/ijerph17051667

Davids, K., Araújo, D., Correia, V., & Vilar, L. (2013). How small-sided and conditioned games enhance acquisition of movement and decision-making skills. *Exercise and*

Sport Sciences Reviews, 41(3), 154–161. https://doi.org/10.1097/JES.0b013e318292f3ec

Drust, B., Reilly, T., & Cable, N. T. (2000). Physiological responses to laboratory-based soccer-specific intermittent and continuous exercise. *Journal of Sports Sciences*. https://doi.org/10.1080/026404100750017814

Durand-Bush, N., & Salmela, J. H. (2002). The development and maintenance of expert athletic performance: Perceptions of world and olympic champions. *Journal of Applied Sport Psychology, 14*(3), 154–171. https://doi.org/10.1080/10413200290103473

Fardilha, F. de S., & Allen, J. B. (2019). Defining, assessing, and developing creativity in sport: A systematic narrative review. *International Review of Sport and Exercise Psychology, 13*(1), 104–127. https://doi.org/10.1080/1750984X.2019.1616315

Glăveanu, V. P. (2013). Rewriting the language of creativity: The five A's framework. *Review of General Psychology, 17*(1), 69–81. https://doi.org/10.1037/A0029528

Greco, P., Memmert, D., & Morales, J. C. P. (2010). The effect of deliberate play on tactical performance in basketball. *Perceptual and Motor Skills*. https://doi.org/10.2466/pms.110.3.849-856

Hendry, D. T., Williams, A. M., & Hodges, N. J. (2018). Coach ratings of skills and their relations to practice, play and successful transitions from youth-elite to adult-professional status in soccer. *Journal of Sports Sciences, 36*(17), 2009–2017. https://doi.org/10.1080/02640414.2018.1432236

Hristovski, R., Davids, K., Araujo, D., & Passos, P. (2011). Constraints-induced emergence of functional novelty in complex neurobiological systems: A basis for creativity in sport. *Nonlinear Dynamics, Psychology, and Life Sciences, 15*(2), 175–206.

Huijgen, B. C. H., Elferink-Gemser, M. T., Ali, A., & Visscher, C. (2013). Soccer skill development in talented players. *International Journal of Sports Medicine, 34*(8), 720–726. https://doi.org/10.1055/s-0032-1323781

Kaufman, J. C., Plucker, J. A., & Baer, J. (2008). *Essentials of creativity assessment*. Wiley.

Kempe, M., & Memmert, D. (2018). "Good, better, creative": The influence of creativity on goal scoring in elite soccer. *Journal of Sports Sciences, 36*(21), 2419–2423. https://doi.org/10.1080/02640414.2018.1459153

Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine, 15*(2), 155. https://doi.org/10.1016/J.JCM.2016.02.012

Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Source: Biometrics, 33*(1), 159–174.

Memmert, D. (2010). Testing of tactical performance in youth elite soccer. *Journal of Sports Science and Medicine, 9*(2), 199–205.

Memmert, D. (2015). Teaching tactical creativity in sport. In *Teaching tactical creativity in sport: Research and practice*. Routledge. https://doi.org/10.4324/9781315797618.

Memmert, D., Baker, J., & Bertsch, C. (2010). Play and practice in the development of sport-specific creativity in team ball sports. *High Ability Studies, 21*(1), 3–18. https://doi.org/10.1080/13598139.2010.488083

Memmert, D., & Roth, K. (2007). The effects of non-specific and specific concepts on tactical creativity in team ball sports. *Journal of Sports Sciences, 25*(12), 1423–1432. https://doi.org/10.1080/02640410601129755

Olthof, S. B. H., Frencken, W. G. P., & Lemmink, K. A. P. M. (2015). The older, the wider: On-field tactical behavior of elite-standard youth soccer players in small-sided games. *Human Movement Science, 41*, 92–102. https://doi.org/10.1016/j.humov.2015.02.004

Orangi, B. M., Yaali, R., Bahram, A., van der Kamp, J., & Aghdasi, M. T. (2021). The effects of linear, nonlinear, and differential motor learning methods on the emergence of creative action in individual soccer players. *Psychology of Sport and Exercise, 56*, Article 102009. https://doi.org/10.1016/j.psychsport.2021.102009

Orth, D., van der Kamp, J., Memmert, D., & Savelsbergh, G. J. P. (2017). Creative motor actions as emerging from movement variability. *Frontiers in Psychology, 8*. https://doi.org/10.3389/fpsyg.2017.01903

Owen, A. L., Twist, C., & Ford, P. (2004). Small-sided games: The physiological and technical effect of altering pitch size and player numbers. *Insight, 7*(2), 50–53.

Rasmussen, L. J. T., Østergaard, L. D., & Glăveanu, V. P. (2019). Creativity as a developmental resource in sport training activities. *Sport, Education and Society, 24*(5), 491–506. https://doi.org/10.1080/13573322.2017.1403895

Roberts, A. H., Greenwood, D. A., Stanley, M., Humberstone, C., Iredale, F., & Raynor, A. (2019). Coach knowledge in talent identification: A systematic review and meta-synthesis. *Journal of Science and Medicine in Sport, 22*(10), 1163–1172. https://doi.org/10.1016/j.jsams.2019.05.008

Santos, S., Coutinho, D., Gonçalves, B., & Sampaio, J. (2023). How many creatives are enough? Exploring how manipulating the number of creative players in the opposing team impacts footballers' performance during small-sided games. *Human Movement Science, 87*, Article 103043. https://doi.org/10.1016/j.humov.2022.103043

Simonton, D. K. (2003). Scientific creativity as constrained stochastic behavior: The integration of product, person, and process perspectives. *Psychological Bulletin, 129*(4), 475–494. https://doi.org/10.1037/0033-2909.129.4.475

Sternberg, R. J., & Lubart, T. I. (1999). The concept of creativity: Prospects and paradigms. In R. J. Sternberg (Ed.), *Handbook of creativity* (pp. 3–15). Cambridge University Press. https://doi.org/10.1017/CBO9780511807916.003.

Torrents, C., Balagué, N., Ric, Á., & Hristovski, R. (2021). The motor creativity paradox: Constraining to release degrees of freedom. *Psychology of Aesthetics, Creativity, and the Arts, 15*(2), 340–351. https://doi.org/10.1037/aca0000291

Torrents, C., Ric, A., Hristovski, R., Torres-Ronda, L., Vicente, E., & Sampaio, J. (2016). Emergence of exploratory, technical and tactical behavior in small-sided soccer games when manipulating the number of teammates and opponents. *PLoS One, 11*(12), Article 168866. https://doi.org/10.1371/journal.pone.0168866

Westmeyer, H. (1998). The social construction and psychological assessment of creativity. *High Ability Studies, 9*(1), 11–21. https://doi.org/10.1080/1359813980090102

Withagen, R., & van der Kamp, J. (2018). An ecological approach to creativity in making. *New Ideas in Psychology, 49*(June 2017), 1–6. https://doi.org/10.1016/j.newideapsych.2017.11.002

Zahno, S., & Hossner, E. J. (2022). Creative actions in team sports are rooted in motor skills rather than in a divergent thinking ability. *German Journal of Exercise and Sport Research*. https://doi.org/10.1007/s12662-022-00847-6

Zahno, S., & van der Kamp, J. (2022). When insights based on ecological and cognitive theories to movement science converge—the case of creativity in sports. *Frontiers in Psychology, 13*. https://doi.org/10.3389/fpsyg.2022.959599